

PRINCIPAL-COMPONENT-BASED GAUSSIAN AFFINE TERM STRUCTURE MODELS: CONSTRAINTS AND THEIR FINANCIAL IMPLICATIONS

RICCARDO REBONATO*, IVAN SAROKA
and VLAD PUTIATYN

*EDHEC Business School — London 10, Fleet Place
Oxford University, OCIAM, St Giles
Morgan Stanley, 25 Cabot Square, UK*

Received 1 May 2019

Accepted 22 December 2019

Published 12 March 2020

This work builds on the work by Joslin *et al.* [(2011) A new perspective on Gaussian dynamic term structure Models, *The Review of Financial Studies* **24**, 926–970] on the affine dynamics of portfolios of yields and addresses the unresolved issues of ‘internal consistency’ mentioned in the same paper. It shows the unexpected constraints that have to be satisfied by the $\mathbb{Q}$ -measure evolution of the yield curve if the portfolio of yields has to be interpreted as their principal components. This choice of state variables is common in the recent literature and so our findings are intrinsically interesting. However, we show that our results also extend to a wide class of choices for state variables, when these are chosen as linear combinations of yields. We show that these constraints have important financial consequences, which, to our knowledge, have not been appreciated. In particular, this paper highlights some puzzling issues of compatibility between the $\mathbb{Q}$ - and $\mathbb{P}$ -measure dynamics of Gaussian dynamic term structure models when principal components are chosen as state variables, once the constraints we derive are taken into account.

Keywords: Affine term structure models; principal components; risk premia; market price of risk.

1. The Goal of This Paper

In this paper, we intend to show that when principal components (or, more generally, what we call ‘specified’ combinations of yields) are used as state variables for Gaussian dynamic term structure models, very strong restrictions must be imposed on their $\mathbb{Q}$ -measure drift. We intend to show (i) that our findings have wide applicability (they extend beyond principal components); and (ii) that they have deep (and disturbing) financial implications about the nature of the market price of risk

*Corresponding author.

implied by these models when these restrictions are combined with empirical findings about excess returns.

These findings are very topical. Indeed, there has recently been a renewed interest in Gaussian dynamic term structure models where the state variables are some linear combinations (‘portfolios’, $\mathcal{P}$) of n yields. See, e.g. Adrian *et al.* (2013). In particular, a number of studies have appeared where the portfolios of yields are the principal components ($\mathcal{X}$) of the yield covariance matrix, or some proxies of the same quantities. See, e.g. Adrian *et al.* (2013) again, Diebold & Rudebusch (2013), and Joslin *et al.* (2011), among others. Furthermore, although modern formulations of affine term structure models use latent (unobservable) state variables (see, e.g. Dai & Singleton 2000 or Duffee 2002), thanks to the results in Duffie & Kan (1996) [see Eq. (5.3)] one can always normalize a model so that the state variables are portfolios of yields — and these portfolios can always be taken to be principal components.

For this class of models, Joslin *et al.* (2011) recently proved that the $\mathbb{Q}$ -measure Gaussian affine dynamics of *any* portfolio of yields, $\mathcal{P}$, are fully characterized by (i) the covariance matrix of the state variables, (ii) the eigenvalues, $\{l_i\}$, of the reversion speed matrix, and (iii) the long-term mean of the short rate under $\mathbb{Q}$. As this holds for any portfolio of yields, $\mathcal{P}$, this is also true for principal-component-based Gaussian affine models. Indeed, we recover this result via a different route, and we show that the eigenvalues, $\{l_i\}$, play a particularly important role when the state variables are principal components.

When Joslin *et al.* (2011) introduced their canonical formulation, they made an important remark that, we believe, has not received the attention it deserves. They wrote (p. 927):

For any N -factor model with portfolio of yields $\mathcal{P}$ as factors, bond prices depend on the $N(N + 1)$ parameters governing the risk-neutral conditional mean of $\mathcal{P}$ and the $N + 1$ parameters linking the short rate to $\mathcal{P}$ [...]. Not all of these parameters are free, however, *because internal consistency requires that the model-implied yields reproduce the yield factors $\mathcal{P}$.* [emphasis added]

They also add in their footnote 9:

Adrian & Moench (2008) explore the case where the pricing factors are the portfolios themselves; *however they do not impose the internal-consistency condition to make the factors equal to their no-arbitrage equivalents...* [emphasis added]

One of the contributions of our paper consists in making these internal consistency conditions explicit in the case when the portfolio of yields is a set of principal components.

As we shall show, these consistency constraints, taken together with well-established empirical information about risk premia, create a tension between the $\mathbb{P}$

and $\mathbb{Q}$ dynamics of yields. More precisely, we show that these models can only support restricted types of (essentially affine) market prices of risk. As it happens, the structure of admissible market prices of risk seems to bear little resemblance with what empirical evidence suggests the dependence of risk premia on level and slope should be. It is for this reason that our restrictions are not just ‘technical’ in nature, but have important financial implications about the ability of Gaussian affine term structure models to capture jointly and realistically the $\mathbb{P}$ and $\mathbb{Q}$ evolution of yields.

One may think that the restrictions we derive are due to the mathematically very ‘special’ nature of principal components, which are obtained by ‘rotating’ yields via special orthogonal matrices. This is far from being the case. Restrictions similar to the ones we find must apply to all choices of state variables that we call ‘specified’: we define precisely what ‘specified’ variables are in what follows, but for the moment, we can think of them as linear combinations with fixed coefficients of yields such as the slope, the average level, the curvature, or, indeed, the principal components. In order to see the essence of the arguments in all its clarity, we explain in the next section the general nature of the restrictions for a particularly simple case.

In the light of these considerations, the paper is organized as follows. In Sec. 2, we explain why the restrictions we derive for the important but specific case of principal components as state variables apply to a very wide class of Gaussian affine term structure models. Then, in Sec. 3, we explain the nature of the ‘tension’ generated by these restrictions between (i) the $\mathbb{Q}$ evolution of yields, (ii) what we empirically know about the yield dependence of the market price of risk, and (iii) the statistically observable $\mathbb{P}$ dynamics of yields. In Sec. 4, we formalize the problem, and we present the constraints in Sec. 5. We show in Sec. 6 what these constraints entail for the $\mathbb{Q}$ -measure evolution of the yield curve. Section 7 discusses how well principal-component-based affine models can recover the cross-section of prices and yield volatilities. Finally, ‘specified variables’ are defined in Sec. 8, and we summarize our findings in Sec. 9. The proofs of our results can be found in the Appendix.

2. The Generality of the Restrictions

Suppose that we take the average of N yields as the first state variable of a dynamic Gaussian term structure model, and the difference between the 10-year and 1-year yields as the second. [As Diebold & Rudebush (2013) show, these variables are very simple and effective proxies for the first and second principal component, but we do not need this observation for what follows.] To make the reasoning even simpler and clearer, we take the average level to be given by the average of the 1-year and the 10-year yields. All we need is that the coefficients that link (linearly) the yields to the state variables should be fixed. We intend to show in this extremely simple case that similar restrictions to the ones derived in the paper for principal components would still apply.

So, after taking the average and the slope as the two state variables, x_1 and x_2 , we place ourselves in an affine setting — as in the body of the paper. We then have (in matrix notation)

$$dx_t = \kappa(\theta - x_t)dt + Sdz_t. \quad (2.1)$$

As usual, for the model to be affine, the short rate must be an affine function of the state variables, i.e. we must have

$$r_t = k_0 + g_1x_1(t) + g_2x_2(t).$$

It is of course always possible to recover without error the initial values of the average yield and slope, $x_1(0)$ and $x_2(0)$. Recall also that, given the affine setting, yields are given by

$$y_t^T = \alpha_t^T + (\beta_t^T)^T x_t, \quad (2.2)$$

with

$$\alpha_t^T = -\frac{1}{T-t}A_t^T, \quad (2.3)$$

$$\beta_t^T = -\frac{1}{T-t}B_t^T, \quad (2.4)$$

and

$$P_t^T = e^{A_t^T + (B_t^T)^T x_t}. \quad (2.5)$$

Now, consider a future time $t > 0$. Given the interpretation, we have given to the two state variables, they should represent the future average yield level, and the future slope. These same state variables, however, also identify future (time- t) yields, $y_t^T = \alpha_t^T + (\beta_t^T)^T x_t$, from which the future average yield and the future slope can be constructed. For instance, we must have $x_2(t) = y_t^{10} - y_t^1$.

However, in general, the future values of the state variables (which are supposed to represent level and slope, and whose initial value was set to be consistent with this interpretation) will not coincide with the same quantities derived as linear combination of yields with fixed coefficients. This means that the future ‘slope-as-state-variable’ will in general not be equal to ‘slope-as-linear-combination-of-future yields’. This, in its simplest form, is the issue of ‘internal consistency’ in Joslin *et al.* (2011). In order to enforce that the variables we have called average level and slope indeed be identifiable as average level and slope (to ensure, that is, that, for any $t > 0$, $x_1(t) = \frac{1}{2}(y_t^1 + y_t^{10})$ and $x_2(t) = y_t^{10} - y_t^1$), we must therefore invoke restrictions on dynamics of the model.

Of course, the same reasoning would apply to any choice of state variables as fixed-coefficients linear combinations of yields. We have not been able to show the form of these restriction for a general linear specification of state variables as a linear combination of yields — indeed, it is plausible, but not obvious, that such a set of restrictions should always exist. In this paper, we derive the (drift) restrictions in

the important case when the linear transformation turns the yield (changes) into principal components.

This section has dealt with the generality of the results. In the next section, we deal with their economical significance.

3. Implications for the Recovery of an Exogenous Market Price of Risk

Before deriving the main results of this paper, it is useful to explain why they pose important questions about principal-component-based affine term-structure modeling.

For the present discussion, we simply need the result (which we will prove in Sec. 5, but which for the moment we will take as true) that, if the state variables are chosen to be principal components, their reversion speed matrix under $\mathbb{Q}$ must have the form

$$\mathcal{K}^{\mathbb{Q}} = \Omega^T F \Lambda_{\mathcal{K}} F^{-1} \Omega, \quad (3.1)$$

where Ω is the matrix of the eigenvectors of the yields, $\Lambda_{\mathcal{K}}$ is (diagonal) matrix with the eigenvalues, $\{l\}$, from the diagonalization of the $\mathbb{Q}$ -measure reversion-speed matrix, and the matrix F is also a function of the same eigenvalues, $\{l\}$.

The reasoning then proceeds as follows. The dependence of the market price of risk on the state variables is an empirical issue. For simplicity of discussion, we shall assume in the following that the market price of risk depends on the slope of the yield curve (see, e.g. Campbell & Shiller 1991, Fama & Bliss 1987) and that what investors are compensated for is ‘level’ risk (see, e.g. Adrian *et al.* 2013, Cochrane & Piazzesi 2005). To keep the discussion simple, let us also identify the first and second principal components with the ‘level’ and ‘slope’ of the yield curve. We stress that the argument in this section does not hinge on these assumptions: if a more complex return-predicting factor were empirically found to predict excess returns well, the argument would still apply, as long as many principal components (reference yields) were retained as required to capture the potentially-more-complex return-predicting factor.

If we accept these stylized facts as true, and if we use the first three principal components as state variables, the market price of risk vector, λ , should be given by

$$\lambda = \lambda_0 + \Lambda_1 x, \quad (3.2)$$

with

$$\Lambda_1 = \begin{bmatrix} 0 & a & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (3.3)$$

(Only the second column of Λ_1 can be nonzero if the dependence of the market price of risk is only on the slope, and not on the level or curvature; the fact that investors only seek compensation for level risk then sets the elements λ_{22} and λ_{32} to zero as well.)

As it is well known, the reversion-speed matrices in the two measures are linked by

$$\mathcal{K}^{\mathbb{Q}} = \mathcal{K}^{\mathbb{P}} - S\Lambda_1. \quad (3.4)$$

The elements of the reversion-speed matrix in the objective measure, $\mathcal{K}^{\mathbb{P}}$, relating to the first principal component may be difficult to estimate with precision because of the quasi-unit-root nature of the level of rates. The second and third principal components, on the other hand, have clearly identifiable (and rather strong) reversion speeds (see, e.g. Diebold & Rudebusch 2014). In any case, after all the statistical uncertainty bars are taken into account, $\mathcal{K}^{\mathbb{P}}$ is an ‘experimental’ fact — and ‘facts of nature’ are also (i) what investors demand compensation for, and (ii) what factors affect the magnitude and sign of the market price of risk (i.e. the structure of the matrix Λ_1).

If we accept these stylized empirical facts, the matrix $\mathcal{K}^{\mathbb{Q}}$ must have the form

$$\mathcal{K}^{\mathbb{Q}} = \mathcal{K}^{\mathbb{P}} - S\Lambda_1 = \begin{bmatrix} \kappa_{11}^{\mathbb{P}} & \kappa_{12}^{\mathbb{P}} - s_{11}a & \kappa_{13}^{\mathbb{P}} \\ \kappa_{21}^{\mathbb{P}} & \kappa_{22}^{\mathbb{P}} & \kappa_{23}^{\mathbb{P}} \\ \kappa_{31}^{\mathbb{P}} & \kappa_{32}^{\mathbb{P}} & \kappa_{33}^{\mathbb{P}} \end{bmatrix}, \quad (3.5)$$

where the last line follows because, for principal components, the nondiagonal elements of the volatility matrix, S , are zero. (If a different dependence of the market price of risk on the principal components were deemed more appropriate, the reasoning would proceed without change.)

Now, as Joslin *et al.* (2011) show and as we re-derive from a different angle in the paper, under mild technical conditions, the reversion speed matrix, $\mathcal{K}^{\mathbb{Q}}$, is fully characterized by its three (in general n) eigenvalues l_1 , l_2 and l_3 . But this raises an important question: once the constraints in Eq. (3.1) are taken into account, is it always possible to find some combination of the (positive, real) eigenvalues $\{l\}$ such that the reversion speed matrix under $\mathbb{Q}$ will assume the form in Eq. (3.5)?

We have not been able to provide general mathematical results about the attainable structure of the recoverable matrices $\mathcal{K}^{\mathbb{Q}}$, but we find that, even for a number of principal components as low as three, the residual flexibility left by the restriction in Eq. (3.1) is not enough to recover the general shape in Eq. (3.5). The $\mathcal{K}^{\mathbb{P}}$ matrices obtained by adding $S\Lambda_1$ to the most compatible $\mathcal{K}^{\mathbb{Q}}$ matrix we can find do not resemble at all the $\mathcal{K}^{\mathbb{P}}$ matrices that can be estimated statistically. (By ‘most compatible’ we mean the Eq. (3.1)-comptabile- $\mathcal{K}^{\mathbb{Q}}$ matrix that can be made to recover the desired empirical information most closely by varying its eigenvalues.) Or, equivalently, if we try to recover from the statistically determined $\mathcal{K}^{\mathbb{P}}$ matrix and the most compatible $\mathcal{K}^{\mathbb{Q}}$ matrix the market-price-of-risk

matrix, Λ_1 , we do not obtain anything similar to what suggested by excess-return studies.

The argument can be further simplified. Let us assume that investors were risk neutral. In that case, the reversion speed matrices in the two measure would coincide: $\mathcal{K}^{\mathbb{Q}} = \mathcal{K}^{\mathbb{P}}$. However, we show in the paper that the $\mathcal{K}^{\mathbb{Q}}$ -measure reversion speed must satisfy the restrictions in Eq. (3.1). These restrictions, as we shall show, strongly constrain the possible shapes of the reversion speed matrix. If investors are risk-neutral, then whatever restriction we find on the $\mathcal{K}^{\mathbb{Q}}$ -measure reversion speed must also apply to its $\mathcal{K}^{\mathbb{P}}$ equivalent. But there is no *a priori* reason to expect that these highly restricted reversion speed matrices should resemble the statistically-determined $\mathbb{P}$ -measure reversion speed matrix. So, a Gaussian dynamic term structure model with principal components as state variables could not represent the yield evolution in a universe with risk-neutral investors.

Therefore, one important question opened by the results we present in this paper is to what extent affine modeling of principal components (and, as explained above, of ‘specified’ state variables) is compatible, given the constraints of Eq. (3.1), with the ‘facts of nature’ embedded in the exogenous reversion-speed matrix $\mathcal{K}^{\mathbb{P}}$, and the exogenous market-risk matrix Λ_1 , both of which are relatively well-known.

Having explained the generality and economic relevance of the our findings, we now move to the formal presentation of the problem.

4. Statement of the Problem

Consider the following dynamics for the component yields of the $N \times 1$ vector y :

$$dy = \mu_y dt + \sigma dw^{\mathbb{Q}, \mathbb{P}}, \quad (4.1)$$

with $dw^{\mathbb{Q}, \mathbb{P}}$ denoting the vector of Brownian shocks in the real-world ($\mathbb{P}$) or in the risk-neutral ($\mathbb{Q}$) measure, with

$$\sigma = \text{diag}[\sigma_1, \sigma_2, \dots, \sigma_N], \quad (4.2)$$

$$E[dw dw^T] = \rho dt, \quad (4.3)$$

and with the drift term μ_y reflecting the no-arbitrage conditions when $dw^{\mathbb{Q}, \mathbb{P}} = dw^{\mathbb{Q}}$, and the real-world dynamics when $dw^{\mathbb{Q}, \mathbb{P}} = dw^{\mathbb{P}}$.

The covariance matrix of yield changes is given by

$$E[dy dy^T] = \sigma \rho \sigma^T = \sigma_{\text{mkt}} dt. \quad (4.4)$$

The quantity σ_{mkt} is assumed to be an exogenous, known and constant market observable.

The real symmetric matrix σ_{mkt} can always be diagonalized to give

$$\sigma_{\text{mkt}} = \Omega \Lambda \Omega^T, \quad (4.5)$$

with the matrix

$$\Lambda = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_N] \quad (4.6)$$

containing the eigenvalues of σ_{mkt} , and with Ω an orthogonal matrix:

$$\Omega\Omega^T = I, \quad (4.7)$$

containing the eigenvectors of σ_{mkt} . If the exogenous matrix σ_{mkt} is positive definite, all the eigenvalues λ_i are positive.

Given this diagonalization, we have

$$dy_t = \Omega dx_t \quad \text{and} \quad dx_t = \Omega^T dy_t \quad (4.8)$$

and we can therefore define as state variables the principal components, x , given by

$$y_t = \tilde{y} + \Omega x_t, \quad (4.9)$$

where $\tilde{y}$ is a *constant* vector.

We then impose that the principal components, x_t , should follow an Ornstein–Uhlenbeck diffusion of the form

$$dx_t = \mathcal{K}(\theta - x_t)dt + Sdz \quad (4.10)$$

and we now choose the Brownian increments, dz , to be orthogonal:

$$E[dzdz^T] = Idt. \quad (4.11)$$

We refer to $\mathcal{K}$ as the reversion-speed matrix, to S as the diffusion matrix, and to θ as the reversion-level vector. For reasons that will become apparent in the following, we require the matrix $\mathcal{K}$ to be invertible and full rank. [Saroka (2015) shows how the full-rank requirement can be relaxed.] Since we want to interpret the factors, x_t , as principal components, we also require the matrix S to be diagonal:

$$S = \text{diag}[s_1, s_2, \dots, s_N] \quad (4.12)$$

and we impose

$$s_i = \sqrt{\lambda_i}. \quad (4.13)$$

Let the affine mapping between the short rate and the state variables, x_t , be of the following form.

$$r_t = u_r + g^T x_t, \quad (4.14)$$

for some vector g to be determined.

Because of Eq. (4.14), absence of arbitrage then imposes that

$$P_t^T = E^{\mathbb{Q}}[e^{-\int_t^T r_s ds}] = E^{\mathbb{Q}}[e^{-\int_t^T (u_r + g^T x_s) ds}]. \quad (4.15)$$

As it is well known, (see, e.g. Dai & Singleton 2000) the solution to Eq. (4.15) is given by

$$P_t^T = \exp^{A_t^T + (B_t^T)^T x_t}, \quad (4.16)$$

with the vector B_t^T and the scalar A_t^T satisfying the ordinary differential equations (with $\tau = T - t$)

$$\frac{dA_\tau}{d\tau} = -u_r + (B_\tau)^T \mathcal{K} \theta + \frac{1}{2} (B_\tau)^T S S^T B_\tau, \quad (4.17)$$

$$\frac{dB_\tau}{d\tau} = -g - \mathcal{K}^T B_\tau, \quad (4.18)$$

and with boundary conditions

$$B(\tau = 0) = 0, \quad A(\tau = 0) = 0. \quad (4.19)$$

It is easy to derive that the solution for B_τ is given by

$$B_\tau^T = g^T \mathcal{K}^{-1} [e^{-\mathcal{K}\tau} - I]. \quad (4.20)$$

From the no-arbitrage dynamics (4.10), and the solution (4.16), the yields vector, y_t , has the expression

$$y_t = \alpha + \beta^T x_t \quad (4.21)$$

with

$$\alpha = - \begin{bmatrix} \frac{A_{\tau_1}}{\tau_1} \\ \frac{A_{\tau_2}}{\tau_2} \\ \dots \\ \frac{A_{\tau_N}}{\tau_N} \end{bmatrix} \quad (4.22)$$

and

$$\beta^T = - \begin{bmatrix} \frac{(B_{\tau_1})^T}{\tau_1} \\ \frac{(B_{\tau_2})^T}{\tau_2} \\ \dots \\ \frac{(B_{\tau_N})^T}{\tau_N} \end{bmatrix}. \quad (4.23)$$

So far, the (standard) results apply to any choice of state variables made up by portfolios of yields. We now require the portfolios of yields to be principal components. If this is the case, for identifiability of the factors with principal components, Eq. (4.9) must also hold

$$y_t = \tilde{y} + \Omega x_t. \quad (4.24)$$

For Eqs. (4.9) and (4.21) to be compatible for an arbitrary vector x_t , one must therefore have

$$\Omega = \beta^T \quad (4.25)$$

and

$$\tilde{y} = \alpha. \quad (4.26)$$

Since the matrix β contains the reversion-speed matrix, $\mathcal{K}$, it is clear that the relationship (4.25) imposes constraints on $\mathcal{K}$. Section 5 derives these constraints.

5. Constraints on the Reversion-Speed Matrix, $\mathcal{K}$

We have obtained that

$$\Omega = \beta^T. \quad (5.1)$$

However, if the state variables are principal components, we must also have

$$\Omega \Omega^T = I, \quad (5.2)$$

and therefore

$$\Omega \beta = I, \quad \beta \beta^T = I. \quad (5.3)$$

The constraints on the reversion speed matrix stem from this condition.

5.1. Impossibility of diagonal $\mathcal{K}^{\mathbb{Q}}$

We start by showing that, under $\mathbb{Q}$, the reversion speed matrix cannot be diagonal. There is no *a priori* theoretical or empirical reason why it should be so. However, looking at this special case gives a clear indication of the nature of the constraints.

As shown, one must have

$$\beta \beta^T = I. \quad (5.4)$$

To prove that the reversion-speed matrix $\mathcal{K}^{\mathbb{Q}}$ cannot be diagonal when the state variables are principal components, we proceed by *reductio ad absurdum*.

So, let us assume that $\mathcal{K}^{\mathbb{Q}}$ is diagonal (and therefore the dynamics of the principal-component state variables decouple). If $\mathcal{K}^{\mathbb{Q}}$ is diagonal, for dynamic stability it is necessary that, for any i , $\kappa_{ii} > 0$. However, even if some κ_{ii} were negative, the ratio $\frac{1 - e^{-\kappa_{ii}\tau k}}{\kappa_{ii}}$ would always be positive. In this setting, it is then straightforward

to show that the matrix β is given by

$$\beta = \begin{bmatrix} \frac{1}{\tau_1} \left[\frac{1 - e^{-\kappa_{11}\tau_1}}{\kappa_{11}} g_1, \frac{1 - e^{-\kappa_{22}\tau_1}}{\kappa_{22}} g_2, \dots, \frac{1 - e^{-\kappa_{NN}\tau_1}}{\kappa_{NN}} g_N \right] \\ \frac{1}{\tau_2} \left[\frac{1 - e^{-\kappa_{11}\tau_2}}{\kappa_{11}} g_1, \frac{1 - e^{-\kappa_{22}\tau_2}}{\kappa_{22}} g_2, \dots, \frac{1 - e^{-\kappa_{NN}\tau_2}}{\kappa_{NN}} g_N \right] \\ \dots \\ \frac{1}{\tau_N} \left[\frac{1 - e^{-\kappa_{11}\tau_N}}{\kappa_{11}} g_1, \frac{1 - e^{-\kappa_{22}\tau_N}}{\kappa_{22}} g_2, \dots, \frac{1 - e^{-\kappa_{NN}\tau_N}}{\kappa_{NN}} g_N \right] \end{bmatrix}, \quad (5.5)$$

with g_j the j th element of the $N \times 1$ vector g in Eq. (4.14). (We note that, given the decoupling of the variables in this setting, each term $\frac{1 - e^{-\kappa_{ii}\tau_j}}{\kappa_{ii}}$ is simply a ‘Vasicek’-like term.) For compactness, we write $[\beta]_{i,j} = h_{ij}g_j$, with $h_{ij} = \frac{1}{\tau_i} \frac{1 - e^{-\kappa_{jj}\tau_i}}{\kappa_{jj}}$.

Consider now $\gamma \equiv \beta\beta^T$. Take a generic off-diagonal element $\gamma[r, s]$, $r \neq s$. For identifiability (i.e. if our state variables were really principal components), one should have

$$[\beta\beta^T]_{rs} = 0, \quad r \neq s. \quad (5.6)$$

In reality, we have

$$[\gamma]_{r,s} = \sum_j \beta_{rj}\beta_{js}^T = \sum_j \beta_{rj}\beta_{sj}. \quad (5.7)$$

But these terms cannot possibly be zero for $r \neq s$, because, for any j , $(g_j)^2 > 0$, and all the terms $h_{js} = \frac{1 - e^{-\kappa_{jj}\tau_s}}{\kappa_{jj}}$ are strictly positive. Therefore, the matrix $\mathcal{K}$ cannot be diagonal.

We can summarize the first result as follows. If the factors x_t are principal components, absence of arbitrage is not compatible with a diagonal reversion-speed matrix, $\mathcal{K}^{\mathbb{Q}}$, under $\mathbb{Q}$.

5.2. Constraints on $\mathcal{K}^{\mathbb{Q}}$

The previous section shows which form the reversion-speed matrix *cannot* assume. To find more constructive constraints on $\mathcal{K}^{\mathbb{Q}}$ in Appendix A, we prove the following two results.

Result 1. Given N yields as above, let the reversion speed matrix, $\mathcal{K}^{\mathbb{Q}}$, be diagonalizable as in

$$\underline{\mathcal{K}^{\mathbb{Q}}} = a\Lambda_{\mathcal{K}}a^{-1}, \quad (5.8)$$

$$\Lambda_{\mathcal{K}} = \text{diag}[l_j], \quad (5.9)$$

with the eigenvalues $\{l_j\}$ distinct, real and positive.^a Let F be the $[N \times N]$ matrix of elements $[F]_{ij}$ given by

$$[F]_{ij} = \frac{1 - e^{-l_{jj}\tau_i}}{l_{jj}\tau_i}. \quad (5.10)$$

Then there always exists a nondiagonal matrix, $\mathcal{K} = \mathcal{K}(l)$, given by

$$\mathcal{K}^{\mathbb{Q}} = \Omega^T F \Lambda_{\mathcal{K}} F^{-1} \Omega, \quad (5.11)$$

such that

$$\Omega \beta^T = I. \quad (5.12)$$

Then, as shown above, if $\Omega \beta^T = I$, the state variables can be interpreted as principal components.

Affine models are characterized by the fact that the short rate (among other quantities) must be an affine function of the state variables. When the state variables are principal components, this uniquely pins down the nature of the affine transformation. More precisely, we show in the Appendix the following result.

Result 2. The affine mapping from the principal-component state variables to the short rate $r_t = u_r + g^T x_t$ must be of the form

$$g = \Omega^T e \quad (5.13)$$

with

$$e = (F^T)^{-1} \mathbf{1}. \quad (5.14)$$

Taken together, the two results uniquely specify (i) the reversion-speed matrix and (ii) the affine mapping from the principal-component state variables to the short rate, once the eigenvalues $\{l_j\}$ are given. In the rest of this paper, we highlight the modeling and financial implication of these constraints on the $\mathbb{Q}$ -measure evolution of the yield curve. Before doing so, however, we observe that, exactly as in Joslin *et al.* (2011), given Ω , the model is still specified once the N eigenvalues of the reversion-speed matrix and one ‘level’ constant are given. The level constant in Joslin *et al.* (2011) in our notation is u_r .

6. What the $\mathbb{Q}$ -Measure Reversion-Speed Matrix Affects

In the approach, we present in this paper, we require the observed yields to be rotated in such a way so as to obtain orthogonal principal components. Once this

^aSaroka (2015) shows that the result can be generalized to the degenerate case when the eigenvalues are real, positive but not distinct. We do not pursue this angle here to keep the argument simple and focused. No conceptual differences arise when using the Jordan normal form of the reversion-speed matrix. Saroka (2015) also deals with the case where the eigenvalues are imaginary, but with positive real part.

‘rotation of axes’ has been made, it permeates every aspect of the resulting mean-reverting $\mathbb{Q}$ -dynamics of principal components. This is because the rotation of axes imposed by the principal-component interpretation of the state variables affects the admissible reversion-speed matrices, which, as we have obtained, must be given by

$$\mathcal{K}^{\mathbb{Q}} = \Omega^T F \Lambda_{\mathcal{K}} F^{-1} \Omega \quad (6.1)$$

for the state variables to be principal components (i.e. for the matrix Ω to be orthogonal).

This link between the reversion speed matrix and the particular rotation singled out by Ω by necessity entails a rather complex mean-reverting deterministic dynamics for the principal components state variables. This can be seen as follows.

First, once the state variables have been chosen to be the principal components, and they have been assigned a mean-reverting behavior as in Eq. (4.10), the conditions of no-arbitrage leave as ‘degrees of freedom’ the N eigenvalues, $\{l\}$, of the reversion speed-matrix. Second, as we have seen, the $\mathcal{K}$ matrix cannot be diagonal. Third, from Eqs. (5.6), (5.7) and (5.12), it follows that some entries of the reversion speed matrix must be negative. Furthermore, the negative entries of the reversion-speed matrix must be large enough to cancel completely the contributions coming from the positive-sign reversion speeds [see Eq. (5.12)]. So, the negative entries of the reversion-speed matrix (which can give rise to a locally mean-fleeing behavior), are not a ‘small correction’, but must be of the same order of magnitude as the positive matrix elements.

It is this feature (and the locally-mean-fleeing behavior between some state variables that it implies) that causes the ‘complex’ deterministic dynamics of the model: these positive and negative reversion speeds simultaneously generate attraction to and significant *repulsion from*, the various state variables and their fixed reversion levels. (For the moment, we call the deterministic dynamics ‘complex’ if, in the absence of stochastic shocks, the expectation of the state variables approaches the relative reversion levels with a nonmonotonic behavior. The larger the amplitude of these oscillations is, and the more numerous the oscillations are, the more complex the resulting behavior. See Figs. 1 and 2.) The overall system *is*, of course, asymptotically stable (because we have required all the eigenvalues l_k to be real and positive^b), but, as shown in detail below, given a set of initial conditions, x_0 , for the state variables, the *deterministic* path to their equilibrium can easily display an oscillatory behavior, with over- and undershoots of their reversion levels. A similar deterministic oscillatory behavior is inherited by the yields, which are affine combination of the factors.

Admittedly, these observations about the oscillatory behavior of the yields (shown in Figs. 1 and 2) refer to their deterministic evolution under $\mathbb{Q}$; however,

^bAs far as stability is concerned, we could allow imaginary eigenvalues with a positive real part. We do not explore this route, for which we see little *a priori* financial justification. This avenue is explored in Saroka (2015).

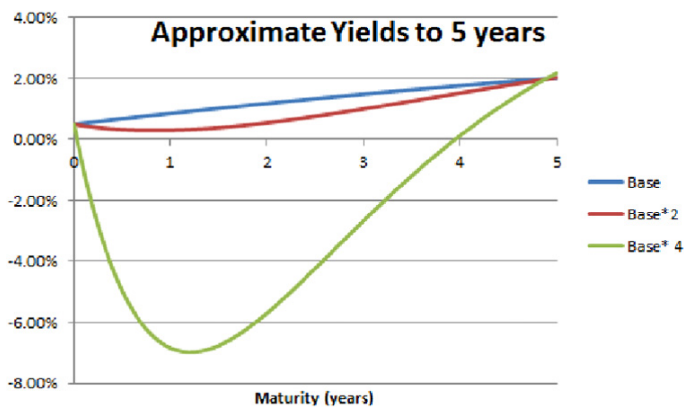


Fig. 1. The averages of the short rate path out to five years for the three cases discussed in the text, namely, for the eigenvalues of the reversion-speed matrix used in the case study (curve labeled “Base”); for eigenvalues twice as large (curve labeled “Base*2”); and for eigenvalues four times as large (curve labeled “Base*4”). In all cases, to within the accuracy of the approximation, the average of the short rate out to five years is indeed 2.00% (the exogenously assigned ‘market’ today value of the five-year yield).

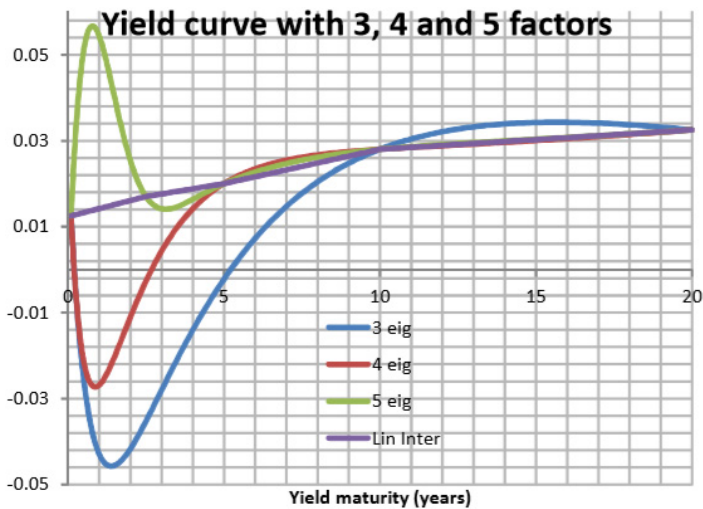


Fig. 2. The same smooth market yield curve described in the text, as recovered by a 3-, 4- and 5-principal-components Gaussian affine model, and a simple linear interpolation. See the text for details.

the only way to avoid a similar behavior for the deterministic evolution of yields under $\mathbb{P}$ would be to have an oscillatory market price of risk, which is similarly implausible. We expand on this point in Sec. 7.

So, we can assign and recover exactly N diagonal elements of the exogenous covariance matrix, and we can assign and recover exactly N elements of an

exogenous (discrete) time-0 yield curve. However, we can only imperfectly specify how the short rate will evolve ‘through the knots’. For large eigenvalues of the reversion-speed matrix (which, we recall, can be arbitrarily assigned under the only constraint that they should be positive and distinct), there can be very large overshoots *and* undershoots, *even if all the N reference ‘market’ yields are exactly recovered* — the larger the trace of the $\underline{K}$ matrix, the ‘wilder’ the over- and under-shootings in between the ‘nodes’ of the market yields.

Figure 1 shows a clear example of this behavior. To understand the three lines in the figure, consider the first yield, which is just the short rate. The time-0 $\mathbb{Q}$ -measure expectation over the paths of the short rate out to a given maturity, T , is straightforwardly related to the time-0 yield of a discount bond of that maturity:

$$y_0^T = -\frac{1}{T} \log E^{\mathbb{Q}}[e^{-\int_0^T r_s ds}]. \quad (6.2)$$

Neglecting for the moment convexity effects (which anyhow plays a significant role only for long maturities), one can approximately write

$$y_0^T \simeq E^{\mathbb{Q}} \left[\frac{1}{T} \int_0^T r_s ds \right]. \quad (6.3)$$

Therefore, simply by averaging out to a horizon T the values of the short rate along a deterministic path, one can immediately (if approximately) relate the path of the short rate to the current model yield curve. At the reference points, by construction, one will observe a perfect match between the market and the model yields (again, within the limits of the approximation above); in between the reference points, however, every choice of eigenvalues will determine, by affecting the path of the state variables to their reversion levels, the values of the intermediate-maturity yields. The reference market yields therefore behave as fixed ‘knots’ through which the yield curve has to go: how smoothly it goes from point-to-point will depend on the eigenvalues of the reversion speed matrix.

This is clearly shown in Fig. 1, where the three lines show the averages of the short rate out to five years for the eigenvalues of the reversion-speed matrix used as the base case for our investigation (top curve labeled “Base”); for eigenvalues twice as large (middle curve labeled “Base*2”); and for eigenvalues four times as large (bottom curve labeled “Base*4”). In all cases, to within the accuracy of the convexity approximation, the average of the short rate out to five years is indeed 2.00% (the exogenously assigned ‘market’ value of y_0^2); however, intermediate yields can assume values which strongly depend on the eigenvalues of the reversion-speed matrix. As mentioned, the larger these eigenvalues, the more ‘complex’ the behavior in between the ‘knots’.

This interpretation also makes more precise the concept of ‘complexity’ for the behavior of yields, to which we have alluded to above: for instance, as in the case of splines, the integral of the curvature of the model yield curve between reference points can be taken as a measure of complexity. So, a possible procedure to choose

the best set of eigenvalues would be to select, among all the yield curves that go exactly through the reference yields, the set of eigenvalues that gives the smallest curvature to the yield curve. We show in Sec. 7 how this smoothness criterion can be effectively used to recover market yields and volatilities.

Similar considerations apply to the covariance matrix. Also, in this case, the choice of the eigenvalues $\{l\}$ strongly influences the values of the ‘interpolated’ covariance elements. Indeed, some choices for the vector l can even produce negative correlations for yields in between the exactly recovered positive covariances. As we shall see in Sec. 7, we will make use of these observations to choose the eigenvalues $\{l\}$.

It is well known that, as one increases the number of principal components, an arbitrary yield curve can be more and more accurately approximated. One may therefore think that one possible solution to the ‘complexity’ problem could be to increase the number of principal components, thereby ‘forcing’ the model yields to recover exactly more and more market yield points. This is alas not the case. As the number of principal components increases, the reversion-speed matrix becomes more and more ‘complex’ in the sense discussed above. As the reversion-speed matrix becomes more ‘complex’, the $\mathbb{Q}$ -expected path of the principal components (and of the yields that can be obtained via rotation) undergoes more and more, and ever more pronounced, undershoots and overshoots before attaining its reversion level. So, indeed, the target yields *are* exactly recovered, but the convergence to the market yield curve is not homogeneous, and the more the target yields (and hence principal components) are added, the wilder the fluctuations *in between* the chosen ‘anchoring’ yields become.

More precisely, Fig. 2 shows the model yields obtained with three, four and five principal components, and the same exogenous reference yield curve described by (i) maturities of 0.1, 10 and 20 years with $y_0^{0.1} = 0.0125$, $y_0^{10} = 0.0280$, and $y_0^{20} = 0.0325$ for three principal components; (ii) maturities of 0.1, 5, 10 and 20 years with $y_0^{0.1} = 0.0125$, $y_0^5 = 0.0200$, $y_0^{10} = 0.0280$, and $y_0^{20} = 0.0325$ for four principal components, and (iii) maturities of 0.1, 2.5, 5, 10 and 20 years with $y_0^{0.1} = 0.0125$, $y_0^{2.5} = 0.0170$, $y_0^5 = 0.0200$, $y_0^{10} = 0.0280$, and $y_0^{20} = 0.0325$ for five principal components.

To keep the comparison as fair as possible, the eigenvalues were chosen to be $[l_1 = 0.2, l_2 = 0.4, l_3 = 0.6]$ for three principal components, $[l_1 = 0.2, l_2 = 0.4, l_3 = 0.6, l_4 = 0.8]$ for four principal components, and $[l_1 = 0.2, l_2 = 0.4, l_3 = 0.6, l_4 = 0.7, l_5 = 0.8]$ for five principal components. We stress that the chosen reference (‘market’) yields are extremely smooth and well-behaved, as the linear interpolation through the yields clearly shows (see the curve labeled ‘Lin Inter’ in Fig. 2). Had the reference yields displayed a less smooth structure, the point illustrated below would have become even more starkly evident.

As the figure clearly shows, the reference yields (three, four and five in the three cases) are always exactly recovered, but, as more points in the yield curve are added, the behavior away from the reference yields becomes wilder and wilder.

For instance, adding the 2.5-year maturity yield (which very ‘naturally’ lies at 1.70% between the 0.1-year yield of 1.25% and the 5-year yield of 2.00%) creates changes in the nonfitted intermediate yields around the 1-year maturity as large as 900 basis points.

In sum: for any number of factors (and hence for any number of exactly recovered market yields), choosing the eigenvalues in such a way so as to obtain as smooth a behavior as possible is always advisable. However, as one imposes the recovery of more and more market yields, it becomes increasingly difficult to control the behavior of the ‘intermediate’ yields. This is because the constraints that we have to impose on the reversion-speed matrix for the interpretation of the variables as principal components to hold generates a behavior for the yields that can become more and more ‘complex’ as one attempts to recover more and more market yields.

In the light of these considerations, we explore in the next section to what extent the smoothness criterion can allow the recovery of interpolated and extrapolated market yields and yield volatilities.

7. Calibration to Market Data

By using the minimum-complexity criterion mentioned in Sec. 6, we have carried out a calibration of the 3-factor version of the model to the discount bond data provided in Gurkaynak *et al.* (2007). By this, we mean that we have first determined the eigenvalues of the reversion speed matrix that give rise to the smoothest overall behavior. Next, keeping these coefficients fixed throughout the sample, we have varied on each trading day the initial values of the state variables so as to recover as well as possible the totality of market yield curves from 1990 to 2014. We also estimated the covariance matrices for the same period.

The results we obtained were always very good, as Figs. 3 and 4 clearly show. In particular, Fig. 4 displays the market and fitted zero-coupon discount yields for a number of representative yield curve shapes observed in the period 1990 to 2014. The fit is clearly excellent, especially considering that the same parameters were used for all the dates, and only the initial values of the state variables were changed.

Similarly, Fig. 3 shows for the statistically estimated and fitted annualized yield volatilities. The full data sample spanning the 1990–2014 period was used for the statistical estimation of the covariance matrix. In this case, we stress that the covariance matrix information fed into the model only extended out to 10 years, and the yield volatilities beyond 10 years are therefore extrapolated by the model. The agreement, again, is excellent.

Encouraging as these results are, we do not dwell on them at length, because, as Diebold & Rudebusch (2013) discuss, we regard dynamic Gaussian term structure models not as yield-curve-fitting tools, but as conveyors of information about the contributions to the yield curve coming from expectation and risk premia. [Variations of the Kim & Wright (2005) affine model, for instance, are routinely used by the Fed to estimate the risk premium for Treasuries.] If one were only interested in the

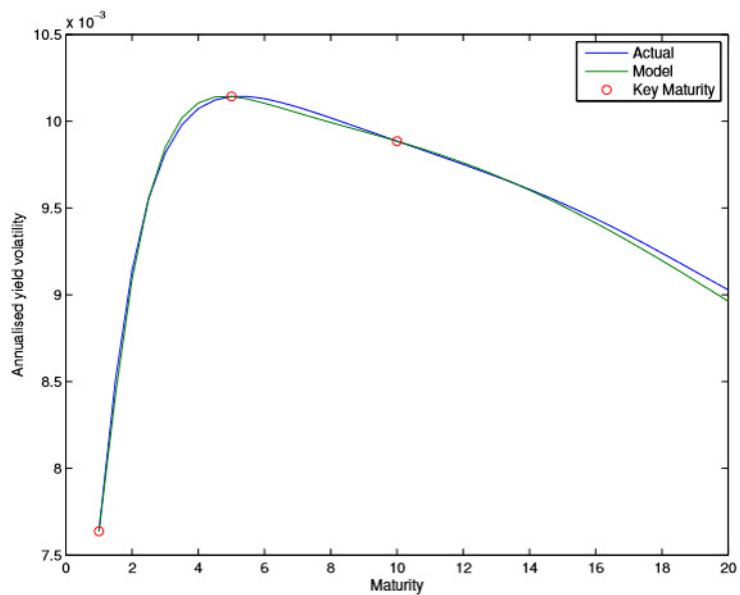


Fig. 3. The statistically estimated and fitted annualized yield volatilities. The full data sample spanning the 1990–2014 period was used for the statistical estimation of the covariance matrix. Yield volatilities beyond 10 years are extrapolated by the model.

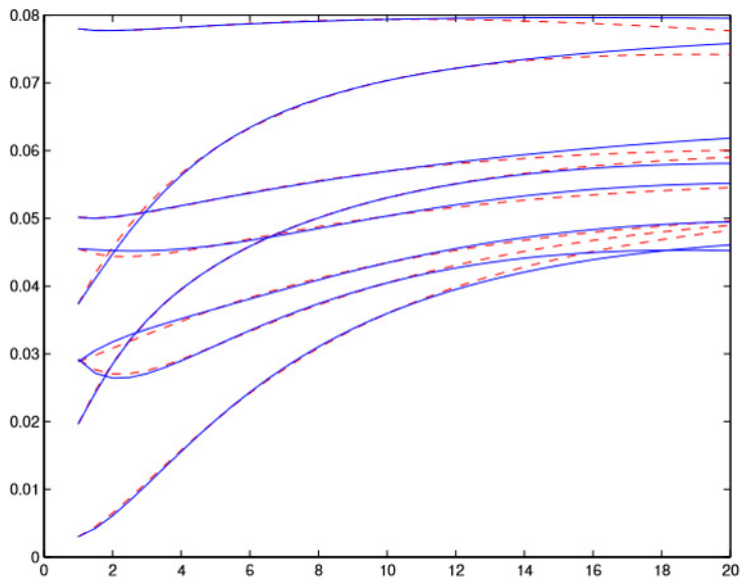


Fig. 4. The market (red dashes) and fitted (continuous blue line) zero-coupon discount yields for a number of representative yield curve shapes observed in the period 1990 to 2014. The same parameters were used for all the fits.

precise recovery of market yields, other approaches (such as the many variations on the Nelson–Siegel 1987 model) can provide as good, or better, fits. Unfortunately, the results we have presented in this paper make us doubt the reliability of the implied market-price-of-risk matrix. More precisely, when we ‘subtract’ from the statistically estimated reversion-speed matrix under $\mathbb{Q}$ the optimal reversion-speed matrix under $\mathbb{Q}$ using

$$\mathcal{K}^{\mathbb{Q}} = \mathcal{K}^{\mathbb{P}} - S\Lambda_1, \quad (7.1)$$

we never find a market-price of risk matrix, Λ_1 , at all similar to what one can expect from the well-established results mentioned in Sec. 3 and reported, *inter alia* in Adrian *et al.* (2013), Cochrane & Piazzesi (2005).

Conversely, when we add to the reversion-speed matrix under $\mathbb{Q}$ reasonable estimates for the term $S\Lambda_1$, no combination of eigenvalues, $\{l_i\}$, allows us to recover a reversion-speed matrix under $\mathbb{P}$ at all similar to what can be easily estimated statistically (see, e.g. Diebold & Rudebusch 2013, Rebonato 2018).

So, impressive as the fits are, we do not have confidence that a principal-component-based dynamic Gaussian term structure model can give reliable information about the joint $\mathbb{Q}$ and $\mathbb{P}$ dynamics. As in the recent literature, these models have been used exactly for this purpose, this should give pause for thought.

8. Defining ‘Specified’ Variables

We stated in Sec. 2 that our results apply to a wide class of state variables, which we labeled as ‘specified’. We conclude this paper by giving a precise definition of the term.

Consider the set of state variables obtained as nontrivial affine combinations of yields with constant coefficients. (We call ‘trivial’ a linear combination with all weights equal to 1 — in which case, of course, we are still dealing with yields.) After evolving the state variables under $\mathbb{Q}$ to a generic future time τ , all the chosen linear combinations of yields must be recovered (i) as time- τ values of the state variables, and (ii) as affine combinations of the time- τ yields. As the yields are, in turn, affine combinations of state variables, this raises general self-consistency issues, as discussed in Sec. 2.

Examples of ‘specified’ state variables could be the slope, curvature or level of the yield curve, a ‘consol’ yield, etc. Examples of nonspecified variables are the yields themselves, latent variables, or macrofinancial variables [such as inflation in Kim & Wright (2005)].

9. Conclusion

With this work, we have tried to contribute to the literature on dynamic Gaussian term structure modeling in several respects. First, we have addressed the issues of ‘internal consistency’ condition mentioned Joslin *et al.* (2011) required ‘to make the

[chosen] factors equal to their no-arbitrage equivalents.’ To our knowledge, these consistency issues had so far been neglected in the literature.

When we enforce these consistency requirements, we find that if principal components (or, more generally, ‘specified-variables’) are chosen as state variables for dynamic Gaussian term structure models, they generate very strong constraints on their reversion-speed matrices. This is our first, and most ‘technical’ result. However, we argue in Sec. 2 that our findings are not due to the very special rotation of yields generated by the principal components, but apply, in different but conceptually similar guises, to all ‘specified-variable’ Gaussian dynamic term structure models.

Despite the seemingly ‘technical’ nature of our results, we have argued that these neglected internal-consistency conditions have important financial implications. Indeed, we have shown that the constraints we derive raise important questions about the ability of principal-component-based affine models to recover the well-established empirical facts about excess returns and the $\mathbb{P}$ -measure reversion-speed matrix: since the $\mathbb{P}$ - and $\mathbb{Q}$ -measure reversion-speed matrices are linked by $\mathcal{K}^{\mathbb{Q}} = \mathcal{K}^{\mathbb{P}} - S\Lambda_1$, what we empirically know about S and λ_1 does not seem to be reconcilable with the restriction on $\mathcal{K}^{\mathbb{Q}}$ imposed by the identification of the state variables as principal components.

Furthermore, we have shown that the constraints we have derived also give rise to a ‘complex’ dynamic behavior under $\mathbb{Q}$. The degree of complexity is dictated by the eigenvalues of the reversion-speed matrix and has a pervasive effect on several market quantities (for instance, it influences the ‘pricing errors’ for the nonreference yields, and the recovery of the nonreference covariance elements). We have suggested a smoothness criterion to choose the values of the eigenvalues.

Finally, we have found that, once the smoothness criterion in discussed in Sec. 6 is employed, a principal-components-based Gaussian term structure model can recover yield curves and yield volatilities extremely well. However, for the reasons explained in detail in Sec. 7, we doubt that the model can be used to give reliable information about the joint $\mathbb{P}$ and $\mathbb{Q}$ dynamics of yields — which is exactly the use it has been often been put to in the recent literature. The considerations presented in Sec. 2 make us fear that similar problems will be encountered for all the ‘specified-variable’ Gaussian term structure models.

Appendix A. Proof of the Constraints on the Reversion-Speed Matrix $\mathcal{K}^{\mathbb{Q}}$

A.1. Preliminaries

Let the affine mapping between the short rate and the state variables, x_t , be of the form in Eq. (4.14)^c:

$$r_t = u_r + g^T x_t. \quad (\text{A.1})$$

^cThis derivation was provided by Dr. Lyashenko (2015, private communication). It is both simpler and more general than the derivation originally arrived at by the authors.

With no loss of generality, we express the vector g as a function of the matrix of the eigenvectors of the yield covariance matrix, $\underline{\Omega}$, as

$$g = \Omega^T e \quad (\text{A.2})$$

for some vector e to be determined. As we have seen, if the state variables then follow a joint dynamics of the form

$$dx_t = \mathcal{K}(\vec{\theta} - x_t)dt + Sdz^{\mathbb{Q}} \quad (\text{A.3})$$

bond prices are given by

$$P_t^T = e^{A_t^T + (B_t^T)^T x_t},$$

where

$$(B_{\tau_i})^T = g^T \mathcal{K}^{-1} [e^{-\mathcal{K}\tau_i} - I] \quad (\text{A.4})$$

is a $1 \times N$ row vector. (To lighten notation, in the following we write $\mathcal{K}$ instead of $\mathcal{K}^{\mathbb{Q}}$.)

Next, we define the time-dependent row vectors β_i^T , and stack them vertically to obtain the $N \times N$ matrix, β^T . The i th element of this matrix is given by

$$\beta_i^T = -\frac{(B_{\tau_i})^T}{\tau_i} = \begin{bmatrix} -\frac{(B_{\tau_1})^T}{\tau_1} \\ -\frac{(B_{\tau_2})^T}{\tau_2} \\ \dots \\ -\frac{(B_{\tau_N})^T}{\tau_N} \end{bmatrix} = -g^T \mathcal{K}^{-1} \frac{1}{\tau_i} [e^{-\mathcal{K}\tau_i} - I]. \quad (\text{A.5})$$

From the body of the paper, we know that, for identifiability of the chosen state variables as principal components, we have to impose

$$\beta \vec{\beta}^T = I. \quad (\text{A.6})$$

Recalling that for the (orthogonal) matrix of the eigenvectors of the yield covariance matrix, Ω , we have

$$\Omega^T \Omega = I, \quad (\text{A.7})$$

it follows that we must have

$$\Omega^T = \beta. \quad (\text{A.8})$$

This is the equation that we set out to prove, i.e. that the generally time-dependent matrix $\beta^T(t)$ (arising from the dynamic of the state variables, x_t) can be constructed in a way that it equals the *constant* matrix Ω (arising from the identification of the

same state variables, x_t , with principal components). To do so, after multiplying both sides of (A.8) by Ω , we obtain

$$\Omega\Omega^T = I = \Omega\beta \quad (\text{A.9})$$

$$= \begin{bmatrix} \vec{\xi}_1 & \vec{\xi}_2 & \cdots & \vec{\xi}_N \end{bmatrix}, \quad (\text{A.10})$$

with

$$\vec{\xi}_j = \Omega \underbrace{\left(-\frac{1}{\tau_j} (\mathcal{K}^T)^{-1} [e^{-\mathcal{K}\tau_i} - I_n] g \right)}_{\vec{\beta}} = -\frac{1}{\tau_j} \Omega (\mathcal{K})^{-1} [e^{-\mathcal{K}\tau_i} - I_n] \Omega^T e, \quad (\text{A.11})$$

where the last line follows from Eq. (A.2). We want to find a reversion speed matrix, $\mathcal{K}$, such that Eq. (A.9) is satisfied.

A.2. Some ancillary results

For any $N \times 1$ vector e

$$e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{bmatrix} \quad (\text{A.12})$$

define

$$\text{Diag}[e] = \begin{bmatrix} e_1 & 0 & \cdots & 0 \\ 0 & e_2 & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & e_N \end{bmatrix}. \quad (\text{A.13})$$

With this definition, it is immediate to show that, for any two e vectors and v , one has

$$\text{Diag}[e]v = \text{Diag}[v]e. \quad (\text{A.14})$$

From this, it follows that, for any $N \times N$ matrices A , B , C and for any vector e one has

$$\begin{bmatrix} A\text{Diag}(B_{j1})Ce & A\text{Diag}(B_{j2})Ce & \cdots & A\text{Diag}(B_{jN})Ce \end{bmatrix} = A\text{Diag}[Ce]B.$$

A.3. Derivation of Result 1 — Restrictions on the matrix $\mathcal{K}$

We are going to apply these results to Eq. (A.11). Before doing so, however, we use the assumption that the reversion-speed matrix is diagonalizable

$$\mathcal{K}^T = b\mathcal{L}b^{-1}. \quad (\text{A.15})$$

With this, we get

$$\begin{aligned}
I &= \begin{bmatrix} \vec{\xi}_1 & \vec{\xi}_2 & \dots & \vec{\xi}_N \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{\tau_1} \Omega(\mathcal{K}^T)^{-1} [I - e^{-\mathcal{K}\tau_1}] \Omega^T e, & \frac{1}{\tau_2} \Omega(\mathcal{K}^T)^{-1} [I - e^{-\mathcal{K}\tau_2}] \Omega^T e, & \dots, \\ \frac{1}{\tau_N} \Omega(\mathcal{K}^T)^{-1} [I - e^{-\mathcal{K}\tau_N}] \Omega^T e \end{bmatrix} \\
&= \begin{bmatrix} \Omega b \text{Diag} \left[\frac{1 - e^{-l_{jj}\tau_1}}{l_{jj}\tau_1} \right] b^{-1} \Omega^T \vec{e}, & \Omega b \text{Diag} \left[\frac{1 - e^{-l_{jj}\tau_2}}{l_{jj}\tau_2} \right] b^{-1} \Omega^T \vec{e}, & \dots, \\ \Omega b \text{Diag} \left[\frac{1 - e^{-l_{jj}\tau_N}}{l_{jj}\tau_N} \right] b^{-1} \Omega^T e \end{bmatrix}.
\end{aligned} \tag{A.16}$$

Using the results obtained above, we can write

$$I = \Omega b \text{Diag}[b^{-1} \Omega^T e] F^T, \tag{A.17}$$

with

$$F[i, j] = \frac{1 - e^{-l_{jj}\tau_i}}{l_{jj}\tau_i}. \tag{A.18}$$

From this, we can solve for the reversion-speed eigenvector matrix b

$$b = \Omega^T (F^T)^{-1} (\text{Diag}[b^{-1} \Omega^T e])^{-1}. \tag{A.19}$$

We can substitute this into Eq. (A.15) to obtain

$$\begin{aligned}
\mathcal{K}^T &= \underbrace{[\Omega^T (F^T)^{-1} (\text{Diag}[b^{-1} \Omega^T e])^{-1}]}_b [\mathcal{L}] \underbrace{[\Omega^T (F^T)^{-1} (\text{Diag}[b^{-1} \Omega^T e])^{-1}]^{-1}}_{b^{-1}} \\
&= \Omega^T (F^T)^{-1} \mathcal{L} F^T \Omega
\end{aligned} \tag{A.20}$$

(in deriving the last line) we have made use of the fact that

$$[\Omega^T (F^T)^{-1} (\text{Diag}[b^{-1} \Omega^T e])^{-1}]^{-1} = \text{Diag}[b^{-1} \Omega^T e] F^T \Omega \tag{A.21}$$

and that $\mathcal{L}$ and commute $\text{Diag}[b^{-1} \Omega^T e]$ (because they are both diagonal).

Finally, this gives

$$\mathcal{K} = \Omega^T \mathbf{F} \mathcal{L} \mathbf{F}^{-1} \Omega, \tag{A.22}$$

which concludes the derivation of the necessary conditions to be satisfied by the reversion-speed matrix if the state variables are principal components.

A.4. Derivation of Result 2 — The vector e

We still need a condition on the vector e in Eq. (A.2)

$$g = \Omega^T e. \quad (\text{A.23})$$

Comparing Eq. (A.22)

$$\mathcal{K} = \Omega^T F \mathcal{L} F^{-1} \Omega, \quad (\text{A.24})$$

and Eq. (A.15)

$$\mathcal{K}^T = b \mathcal{L} b^{-1} \quad (\text{A.25})$$

one finds

$$b = \Omega^T (F^T)^{-1}. \quad (\text{A.26})$$

To get the condition on vector e , substitute this expression for b in Eq. (A.17)

$$\begin{aligned} I &= \Omega b \text{Diag}[b^{-1} \Omega^T e] F^T \\ &= \Omega \Omega^T (F^T)^{-1} \text{Diag}[F^T \Omega \Omega^T e] F^T \\ &\Rightarrow I = (F^T)^{-1} \text{Diag}[F^T e] F^T. \end{aligned} \quad (\text{A.27})$$

Now, multiply from the left by F^T and from the right by $(F^T)^{-1}$. This gives

$$I = \text{Diag}[F^T e]. \quad (\text{A.28})$$

This implies that

$$F^T e = \begin{bmatrix} 1 \\ 1 \\ \dots \\ 1 \end{bmatrix} \equiv 1. \quad (\text{A.29})$$

(The symbol 1 signifies a vector of 1s.) Finally, we have

$$e = (F^T)^{-1} 1. \quad (\text{A.30})$$

This means that in the formulae

$$r_t = u_r + g^T x_t, \quad (\text{A.31})$$

$$g = \Omega^T e, \quad (\text{A.32})$$

one must have

$$e = (F^T)^{-1} 1, \quad (\text{A.33})$$

$$r_t = u_r + \Omega^T (F^T)^{-1} 1 x_t. \quad (\text{A.34})$$

References

- T. Adrian, R. K. Crump & E. Moench (2013) *Pricing the Term Structure with Linear Regressions*, Federal Reserve Bank of New York, Staff Report No. 340, August 2008, Revised 2013, available at http://www.newyorkfed.org/research/staff_reports/sr340.pdf.
- A. Brace, D. Gatarek & M. Musiela (1997) The market model of interest rate dynamics, *Mathematical Finance* **7** (2), 127–147.
- J. Campbell & R. Shiller (1991) Yield spreads and interest rate movements: A bird's eye view, *Review of Economic Studies* **58**, 495–514.
- J. H. Cochrane & M. Piazzesi (2005) Bond risk premia, *American Economic Review* **95** (1), 138–160.
- Q. Dai & K. J. Singleton (2000) Specification analysis of affine term structure models, *Journal of Finance* **55** (5), 1943–1978.
- F. X. Diebold & G. D. Rudebusch (2013) *Yield Curve Modelling and Forecasting — The Dynamic Nelson-Siegel Approach?* Princeton, NJ, and Oxford, UK: Princeton University Press.
- G. Duffee (2002) Term premia and interest rate forecasts in affine models, *Journal of Finance* **57**, 405–443.
- D. Duffie & R. Kan (1996) A yield-factor model of interest rates, *Mathematical Finance* **6**, 379–406.
- E. F. Fama & R. Bliss (1987) The information in long-maturity forward rates, *American Economic Review* **77**, 680–692.
- R. S. Gurkaynak, B. Sack & J. H. Wright (2006) *The US Treasury Yield Curve: 1961 to the Present*, working paper no 2006-28, Federal Reserve Board, Washington, DC, Division of Research, Statistics and Monetary Affairs.
- R. S. Gurkaynak, B. Sack & J. H. Wright (2007) The US treasury yield curve: 1961 to the present, *Journal of Monetary Economics* **54**, 2291–2304, data available at <http://www.federalreserve.gov/econresdata/researchdata/feds200805.xls> and <http://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html>.
- S. Joslin, A. Le & K. J. Singleton (2013) Why Gaussian macro-finance models are (Nearly) unconstrained factor-VARs, *Journal of Financial Economics* **109** (3), 604–622.
- S. Joslin, K. J. Singleton & H. Zhu (2011) A new perspective on Gaussian dynamic term structure models, *The Review of Financial Studies* **24**, 926–970.
- D. H. Kim & J. H. Wright (2005) *An Arbitrage-Free Three-Factor Term Structure Model and Recent Behavior of Long-Term Yields and Distant-Horizon Forward Rates*, Finance and Economics Discussion Series, Division of Statistics and Monetary Affairs, Federal Reserve Board, Washington, DC, 2005–33.
- C. R. Nelson & A. F. Siegel (1987) Parsimonious modeling of yield curves, *Journal of Business* **60**, 473–489.
- R. Rebonato (2018) *Bond Pricing and Yield Curve Modelling: A Structural Approach*, Cambridge, UK: Cambridge University Press.
- I. Saroka (2015) *Arbitrage-Free Modelling of Affine Principal Components of the Yield Curve*, Oxford University, unpublished thesis for MSc in Mathematical Finance.

Copyright of International Journal of Theoretical & Applied Finance is the property of World Scientific Publishing Company and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.